

Partitions

Definition Let A be a set and $\mathcal{A} = \{A_n : n \in I\}$ be a collection of non-empty subsets $A_n \subseteq A$. We say that A is *partitioned by* $\mathcal{A}$ if

1. $A = \bigcup_{n \in I} A_n$. (the A_n together make up A)
2. If $A_m \neq A_n$, then $A_m \cap A_n = \emptyset$. (distinct A_n are pairwise disjoint)

We describe the collection $\mathcal{A}$ as a partition of A .

Note:

Let A be a nonempty set. The family $\mathcal{A}$ is a **partition** of A if and only if

- $\mathcal{A} \subseteq \mathcal{P}(A)$,
- $\bigcup \mathcal{A} = A$,
- $\mathcal{A}$ is pairwise disjoint.

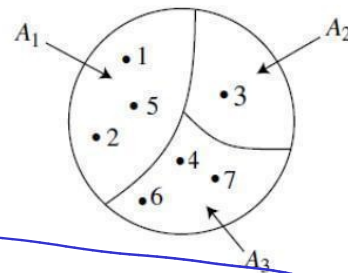
Example:

Let $A = \{1, 2, 3, 4, 5, 6, 7\}$ and define the elements of the partition to be $A_0 = \{1, 2, 5\}$, $A_1 = \{3\}$, and $A_2 = \{4, 6, 7\}$

The family $\mathcal{A} = \{A_1, A_2, A_3\}$ is a subset of $\mathcal{P}(A)$,

$A = A_0 \cup A_1 \cup A_2$, and $\mathcal{A}$ is pairwise disjoint.

Therefore, $\mathcal{A}$ is a partition of A .



note that
 $A_0 \cap A_1 = \emptyset$
 $A_0 \cap A_2 = \emptyset$
 $A_1 \cap A_2 = \emptyset$

Lemma. Let A be a set.

Suppose that $\sim$ is an equivalence relation on A . Then $a \sim b \iff [a] = [b]$.

Note: Distinct equivalence classes are pairwise disjoint: if $[x] \neq [y]$, then $[x] \cap [y] = \emptyset$.

Theorem. Let A be a set.

1. If $\sim$ is an equivalence relation on A , then A is partitioned by the equivalence classes of $\sim$.
2. If $\{A_n\}_{n \in I}$ is a partition of A , then the relation $\sim$ on A defined by

$$a \sim b \iff \exists n \in I \text{ such that } a \in A_n \text{ and } b \in A_n$$

is an equivalence relation.

Partial Order Relation

A relation $\mathcal{R}$ is said to be a partial order relation if it is simultaneously reflexive, antisymmetric, and transitive on A . set on which there is a partial order relation defined is called a *poset*.

Example :

- $\leq$ is a partial order relation on $\mathbb{N}, \mathbb{Z}$ and $\mathbb{R}$.
- The relation $\mathcal{R} = \{(a, b) \mid a \subseteq b\}$ define on the power set of $A = \{1, 2, 3\}$ is a partial order relation.

Remark 1 The word "partial" in partial order indicates that not every pairs of elements in a set is comparable. (two elements a and b of poset A are called comparable if $a\mathcal{R}b$ or $b\mathcal{R}a$).

Example : In the poset, $(\mathcal{P}(A), \subseteq)$, where $A = \{1, 2, 3\}$
 $\{1, 3\} \not\subseteq \{2\}$ and $\{2\} \not\subseteq \{1, 3\}$. Therefore, $\{1, 3\}$ and $\{2\}$ are not comparable. On the other hand, $\{1\} \subseteq \{1, 3\}$ are comparable.

Note:

If a relation $\preccurlyeq$ on a set A is reflexive, antisymmetric, and transitive, $\preccurlyeq$ is a **partial order** on A and the ordered pair $(A, \preccurlyeq)$ is called a **partially ordered set** (or simply a **poset**). Furthermore, for all $a, b \in A$, the notation $a < b$ means $a \preccurlyeq b$ but $a \neq b$.

For example, $\leq$ and $=$ are partial orders on $\mathbf{R}$, but $<$ is not a partial order on $\mathbf{R}$ because the relation $<$ is not reflexive. Although $=$ is a partial order on any set, in general an equivalence relation is not a partial order

Theorem [Weak-Trichotomy Law]

If $\preccurlyeq$ is a partial order on A , for all $a, b \in A$, at most one of the following are true:
 $a < b$, $b < a$, or $a = b$.

Proof:

Let $a, b \in A$. We have three cases to consider.

- Suppose $a < b$. This means that $a \preccurlyeq b$ and $a \neq b$. If in addition $b < a$, by transitivity $a < a$, which is a contradiction.
- That $b < a$ precludes both $a < b$ and $a = b$ is proved like the first case.
- If $a = b$, then by definition of $<$ it is impossible for $a < b$ or $b < a$ to be true. ■

Bounds

Let $\preccurlyeq$ be a partial order on A with elements m and m' such that $a \preccurlyeq m$ and $a \preccurlyeq m'$ for all $a \in A$. In particular, this implies that $m \preccurlyeq m'$ and $m' \preccurlyeq m$, so since $\preccurlyeq$ is antisymmetric, we conclude that $m = m'$. Similarly, if $m \preccurlyeq a$ and $m' \preccurlyeq a$ for all $a \in A$, then $m = m'$. This argument justifies the use of the word *the* in the next definition.

Definition:

Let $(A, \preccurlyeq)$ be a poset and $m \in A$.

- m is the **least element** of A (with respect to $\preccurlyeq$) if $m \preccurlyeq a$ for all $a \in A$.
- m is the **greatest element** of A (with respect to $\preccurlyeq$) if $a \preccurlyeq m$ for all $a \in A$.

There is no guarantee that a partially ordered set will have a least or a greatest element.

Example:

If A is a finite set of real numbers, A has a least and a greatest element with respect to $\leq$. Under the partial order $\leq$, the set $\mathbf{Z}^+$ also has a least element but no greatest element, and both $\{5n : n \in \mathbf{Z}^-\}$ and $\mathbf{Z}^-$ have greatest elements but no least elements. To show that $B = \{5n : n \in \mathbf{Z}^+\}$ has a least element with respect to $\leq$, use the fact that the least element of $\mathbf{Z}^+$ is 1. Therefore, the least element of B is 5 because $5 \in B$ and $5(1) \leq 5n$ for all $n \in \mathbf{Z}^+$.

Definition:

Let $(A, \preccurlyeq)$ be a poset and $a, b \in A$. If $a \preccurlyeq b$ or $b \preccurlyeq a$, then a and b are **comparable** with respect to $\preccurlyeq$. Elements of A are **incomparable** if they are not comparable.

Definition:

Let $(A, \preccurlyeq)$ be a poset and $m \in A$.

- m is a **minimal element** of A (with respect to $\preccurlyeq$) if $a \not\preccurlyeq m$ for all $a \in A$.
- m is a **maximal element** of A (with respect to $\preccurlyeq$) if $m \not\preccurlyeq a$ for all $a \in A$.

Definition:

A subset C of the poset $(A, \preccurlyeq)$ is a **chain** with respect to $\preccurlyeq$ if a is comparable to b for all $a, b \in C$.

Example:

Let $\{A_k : k \in \mathbf{Z}^+\}$, where $A_k = \{x \in \mathbf{Z} : (x-1)(x-2) \cdots (x-k) = 0\}$ is a chain with respect to $\subseteq$, take $m, n \in \mathbf{Z}^+$. By definition,

$$A_m = \{1, 2, \dots, m\}$$

and

$$A_n = \{1, 2, \dots, n\}.$$

If $m \leq n$, then $A_m \subseteq A_n$, otherwise $A_n \subseteq A_m$.

Definition: Linear order (Total order)

The poset $(A, \preccurlyeq)$ is a **linearly ordered set** and $\preccurlyeq$ is a **linear order** if A is a chain with respect to $\preccurlyeq$.

Since every subset A of $\mathbf{R}$ is a chain with respect to $\leq$, the relation $\leq$ is a linear order on A . Furthermore, since every pair of elements in a linear order are comparable,

Definition

Let $(A, \preccurlyeq)$ be a poset. The elements $a, b \in A$ are **compatible** if there exists $c \in A$ such that $c \preccurlyeq a$ and $c \preccurlyeq b$. If a and b are not compatible, they are **incompatible** and we write $a \perp b$.

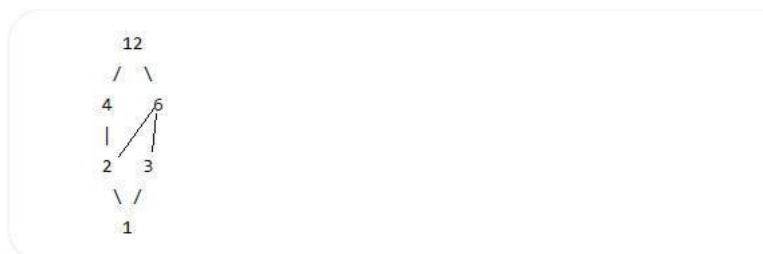
This definition means that a and b have a common lower bound.

Example 1: Divisibility Poset (Compatible)

Let $A = \{1, 2, 3, 4, 6, 12\}$,

ordered by divisibility: $a \leq b \Leftrightarrow a \mid b$.

The Hasse diagram is



Consider $a = 4$ and $b = 6$.

Since $1 \leq 4$ and $1 \leq 6$, the element 1 is a common lower bound.

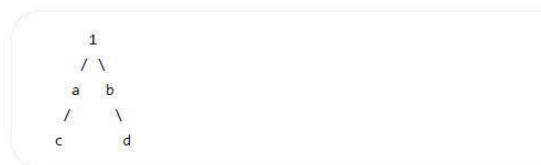
Also, $2 \leq 4$ but $2 \not\leq 6$.

Hence, 4 and 6 are compatible.

However, 4 and 6 are incomparable, since they belong to different chains.

Example 2: A Slightly Larger Poset

Consider



where the order is upward.

- $c \leq a, d \leq b$.
- The elements a and b have no common lower bound.
- Therefore, a and b are incompatible

Definition:

The linearly ordered set $(A, \preceq)$ is a **well-ordered set** and $\preceq$ is a **well-order** if every nonempty subset of A has a least element with respect to $\preceq$.

According to Definition, $(\mathbb{N}, \leq)$ is a well-ordered set, but this fact about the natural numbers cannot be proved without making an assumption. Therefore, so that we have at least one well-ordered set with which to work.

Theorem

If $(A, \preceq)$ is well-ordered and B is a nonempty subset of A , then $(B, \preceq)$ is well-ordered.

Proof:

Let $B \subseteq A$ and $B \neq \emptyset$. To prove that B is well-ordered, let $C \subseteq B$ and $C \neq \emptyset$. Then, $C \subseteq A$. Since $\preceq$ well-orders A , we know that C has a least element with respect to $\preceq$. ■

5 Functions

Basic Definitions

Definition. A function is an ordered triple (f, A, B) such that

1. A and B are sets, and $f \subseteq A \times B$,
2. For every $x \in A$ there is some $y \in B$ such that $(x, y) \in f$
3. If $(x, y) \in f$ and $(x, z) \in f$, then $y = z$; in other words, the assignment is unique in the sense that an $x \in A$ is assigned at most one element of B .

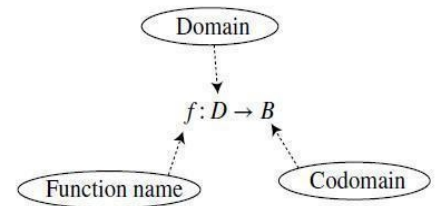
A is called the domain of f , and B its codomain.

It is customary to write the function (f, A, B) as $f : A \rightarrow B$.

Also, if $(x, y) \in f$, then we will usually write $y = f(x)$,

and call y the *image of x under f* .

The set $\{y \in B : \text{There is an } x \in A \text{ such that } y = f(x)\}$ is called the range of f .



Notation

If f is a function from A to B we write $f : A \rightarrow B$.

If $a \in A$, we write $b = f(a)$ for the the element of B assigned to a by the function f .

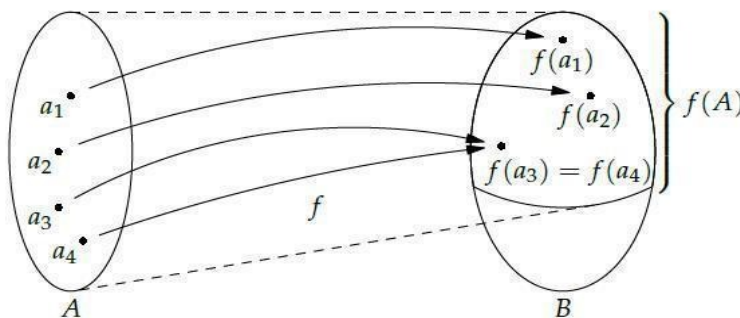
We can also write $f : a \mapsto b$, which is read ' f maps a to b .'

If U is a subset of A then the image of U is the following subset of B ,

$$f(U) = \{f(u) \in B : u \in U\}.$$

The image of A is precisely the range of f , hence the notation $\text{Im}(f)$,

$$f(A) = \text{range}(f) = \text{Im}(f) = \{f(a) : a \in A\}.$$



Remarks(Domain and range of functions):

Let A, B be sets and $f : A \rightarrow B$ be a function.

According to the definition, $D(f) = A$ is the **domain** of the function f and $R(f) = \{f(x) | x \in D(f)\}$ is the **range** of the function f . The set B is called the **co-domain** of the function f . Note that $R(f) \subseteq B$.

Summary

A function consists of **three** parts:

1. **domain**,
2. **codomain**,
3. and a **rule** that describes how to assign a unique range object to each object in the domain.

Example 1:

Show that the relation $f = \{(1, a), (2, b), (3, a)\}$ defines a function from $A = \{1, 2, 3\}$ to $B = \{a, b, c\}$. Find its range.

Solution:

Clearly $f \subseteq A \times B$ and $D(f) = A$.

Since every element of A has a **unique** element in B , f is a function.

The range of f

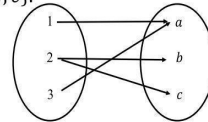
$$\begin{aligned} R(f) &= \{f(x) | x \in D(f)\} \\ &= \{f(x) | x \in \{1, 2, 3\}\} \\ &= \{f(1), f(2), f(3)\} \\ &= \{a, b\} \end{aligned}$$

consists of the elements a and b .

Note that $R(f) \subseteq B$.

Example 2:

Show that the relation $f = \{(1, a), (2, b), (2, c), (3, a)\}$ Does not define a function from $A = \{1, 2, 3\}$ to $B = \{a, b, c\}$.



"uniqueness" violated for 2 appears twice

Solution:

Clearly $f \subseteq A \times B$ and $D(f) = A$.

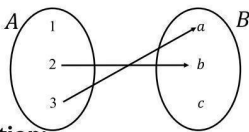
But $2 \in A$ has two images in B .

Therefore, f is not a function from A to B .

Example 3:

Show that the relation $f = \{(2, b), (3, a)\}$

Does not define a function from $A = \{1, 2, 3\}$ to $B = \{a, b, c\}$.



"For each" violated

Solution:

Clearly $f \subseteq A \times B$.

But $D(f) = \{2, 3\} \neq A$.

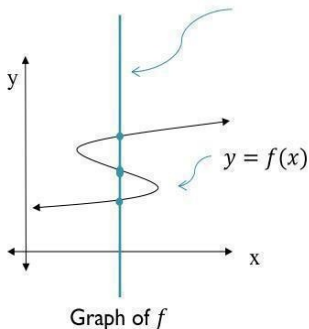
Therefore, f is not a function from A to B .

Graph of function:

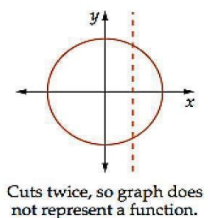
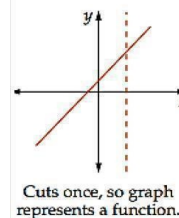
The graph of a function f is the set of all ordered pairs $\{(x, f(x)) | x \in D(f)\}$.

Vertical Line Test:

A curve in the xy -plane is a graph of a function if and only if **every vertical line intersects it at most once**.



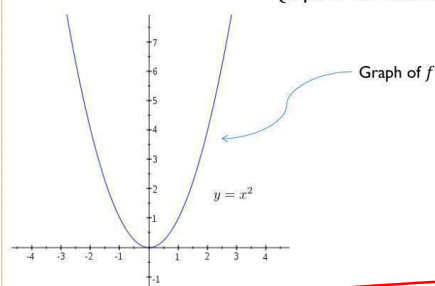
f is not a function
Uniqueness violated for some x 's

**Example 4:**

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2$ for all $x \in \mathbb{R}$.

Then, f is a function. $D(f) = \mathbb{R}$ and

$$\begin{aligned} R(f) &= \{f(x) | x \in \mathbb{R}\} \\ &= \{x^2 | x \in \mathbb{R}\} \\ &= \{x | x \in \mathbb{R} \text{ and } x \geq 0\}. \end{aligned}$$



End of D22

Definition. Two functions $f : A \rightarrow B$ and $g : C \rightarrow D$ are **equal** if and only if

1. $A = C$,
2. $B = D$,
3. $f = g$. ($f = g$ if and only if $f(x) = g(x)$ for every $x \in A$.)

Definition . Let $f : A \rightarrow B$ and $g : C \rightarrow D$ be functions such that $\text{ran } f \subseteq \text{dom } g$. Then the function $g \circ f : A \rightarrow D$ defined by $(g \circ f)(x) = g(f(x))$ is called the *composition* of f and g .

Note:

$$D(g \circ f) = \{x \mid x \in D(f) \text{ and } f(x) \in D(g)\}.$$

Lemma1. If f and g are functions such that $\text{ran } f \subseteq \text{dom } g$, then $\text{dom}(g \circ f) = \text{dom } f$, and $\text{codom}(g \circ f) = \text{codom } g$.

Lemma 2. Let $f : A \rightarrow B$, $g : B \rightarrow C$, $h : C \rightarrow D$ be functions. Then, $h \circ (g \circ f) = (h \circ g) \circ f$, i.e. the composition of functions is associative.

Proof. Let $p = h \circ (g \circ f)$, and $q = (h \circ g) \circ f$; to show that two functions are equal.